

FREE RESOURCE | THE MBA PA

Cardiac Surgery First-Assist Protocol Template

A Practical Readiness Framework for Cardiac Surgery PAs, Surgical Techs, and Surgical Residents

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The MBA PA

Helping cardiac surgery teams strengthen onboarding, improve first-assist development, and build training systems that scale.

How to Use This Resource

This template is designed for cardiac surgery teams that want a cleaner, safer, more teachable approach to first-assist performance in the OR. It can be used in three ways:

- **For individual clinicians:** as a self-audit before cases
- **For lead PAs / educators:** as a competency and onboarding framework
- **For programs:** as the starting point for a standardized first-assist protocol

This is not a substitute for attending preference, institutional policy, or direct supervision. Use it to reduce variability, improve preparation, and shorten the learning curve.

1) First-Assist Readiness: Pre-Case Protocol

Before entering the room, the first assistant should be able to answer five basic questions clearly.

A. The Patient Snapshot

Confirm you know:

- Procedure being performed
- Surgical indication
- Relevant anatomy or pathology
- Prior sternotomy or re-entry risk
- Relevant conduit plan (LIMA, RIMA, SVG, radial, none)
- Relevant comorbidities that affect flow of case
 - severe COPD
 - CKD
 - peripheral vascular disease
 - obesity
 - bleeding risk / anticoagulation issues
 - low EF / hemodynamic fragility

Minimum standard: You should be able to give a 30-second case summary without reading directly from the chart.

B. The Procedure Map

Know the expected sequence for the specific operation.

Example sequence: CABG

- Positioning / prep / drape
- Sternotomy
- Conduit harvest / prep
- Pericardiotomy
- Cannulation setup if on-pump
- Cross-clamp / distal anastomoses / proximals
- De-airing / separation from bypass if applicable
- Hemostasis

- Chest tube / wire / closure

Example sequence: Valve / combined case

- Exposure strategy
- Cannulation plan
- Aortotomy or atriotomy access
- Valve exposure and instrumentation needs
- Closure of cardiac incisions
- De-airing / echo checkpoints
- Hemostasis / closure

Minimum standard: You should know what the surgeon will likely need next before they ask.

C. The Exposure Plan

Review in advance:

- Bed position requirements
- Arm positioning / access concerns
- Sternal retractor setup
- Suction setup
- Headlight / loupes / specialty instruments
- Exposure aids likely needed for distal targets, posterior pericardium, valve visualization, or redo dissection

D. The Critical Instruments / Materials List

Identify the items that will create delay if missing.

TYPICAL EXAMPLES:

- Sternal saw backup plan
- Wire passer / closure system
- Bulldog clamps / coronary shunts

- Fine forceps / needle drivers
- Vein markers / conduit clips / vessel loops
- Aortic punch / partial clamp
- Valve sizers / knot pushers / pledgets
- Hemostatic agents and patch materials

E. The Risk Check

Ask yourself:

- What are the two most likely points of difficulty in this case?
- Where could exposure break down?
- What part of the case requires the fastest anticipation from me?
- What does this surgeon hate waiting for?

That last question matters more than people admit.

2) Intraoperative First-Assist Standards

A strong first assistant does more than “help.” They control flow, preserve exposure, reduce friction, and make the surgeon more efficient without creating noise.

A. Exposure

CORE STANDARD:

- Maintain a stable field without drifting, over-pulling, or blocking the surgeon’s line of sight

BEST PRACTICES:

- Retract with intention, not force
- Adjust retraction as tissue tension changes

- Protect conduit, myocardium, phrenic nerve, and friable structures
- Reposition early when exposure is degrading; do not wait until the surgeon is frustrated
- Keep your hands out of the surgeon's working lane whenever possible

Self-check: If exposure is poor, ask whether the issue is angle, tension, depth, suction, light, or sequence.

B. Anticipation

CORE STANDARD:

- Be one step ahead on instruments, sutures, visualization needs, and next movement

BEST PRACTICES:

- Track case progress continuously
- Know surgeon preference for suture type, clamp timing, and handoff rhythm
- Prepare the next step before verbal request whenever safe
- Communicate only when it improves safety, timing, or clarity

Rule: Good anticipation feels quiet. If you are constantly asking what comes next, your preparation happened too late.

C. Tissue Respect

CORE STANDARD:

- Handle tissue as if every unnecessary movement increases bleeding, edema, or repair time

BEST PRACTICES:

- Avoid crushing conduit or fragile tissue
- Use deliberate needle angles and controlled follow-through
- Support structures rather than chasing them

- Keep the field clean enough to distinguish anatomy, not just “less bloody”

D. Hemostasis Support

CORE STANDARD:

- Recognize bleeding sources early and support correction without panic

BEST PRACTICES:

- Differentiate needle-hole bleeding, raw surface oozing, conduit bleeding, and suture-line bleeding
- Improve visualization before offering solutions
- Have suction, sponge, clips, and hemostatic adjuncts ready before asked in high-risk portions of the case
- During closure, slow down enough to actually inspect

E. Communication

CORE STANDARD:

- Speak briefly, clearly, and at the right moment

Use closed-loop communication when it matters:

- “Prolene loaded.”
- “Bulldog on.”
- “Suction coming left side.”
- “Needle back.”
- “Bleeder at inferior edge.”

Avoid:

- Narrating obvious actions
 - Guessing when you are unsure
 - Talking through corrections instead of making them
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3) Common First-Assist Failure Points and Fast Corrections

Problem 1: Exposure keeps collapsing

LIKELY CAUSES:

- Retraction angle is wrong
- Bed or patient position is suboptimal
- Suction is fighting exposure instead of helping it
- Assistant is holding tension statically while anatomy has shifted

FAST CORRECTION:

- Reset hand position
- Reassess depth and direction of pull
- Ask for a brief pause to re-establish field if needed
- Use two-point exposure instead of overcommitting one hand

Problem 2: Surgeon keeps asking for things before you are ready

LIKELY CAUSES:

- Weak procedural map
- Inattention during transitions
- Not learning surgeon-specific patterns

FAST CORRECTION:

- Build a "next three moves" habit
- Before each major step, mentally name the likely next instrument, suture, and exposure need
- After the case, write down 3-5 recurring preference patterns

Problem 3: Your retraction is technically present but not actually useful

LIKELY CAUSES:

- You are pulling harder instead of smarter
- You are blocking motion or visualization
- You are creating tissue trauma for marginal gain

FAST CORRECTION:

- Reduce force
- Change vector
- Improve light and suction first
- Ask yourself what structure truly needs to be seen right now

Problem 4: You lose control during bleeding moments

LIKELY CAUSES:

- Poor visual discipline
- Moving too quickly without identifying source
- Talking too much under stress

FAST CORRECTION:

- Clear field
- Stabilize exposure
- Identify source
- Support surgeon's hands, line of sight, and immediate tool needs
- Keep communication short and literal

Problem 5: Closure becomes rushed and sloppy

LIKELY CAUSES:

- Team relaxes after the “main part” of the case
- Instrument organization deteriorates
- Hemostasis check is incomplete

FAST CORRECTION:

- Treat closure as its own phase, not leftover work
- Reconfirm wires, drains, counts, pacing wires, and line positioning
- Do a final deliberate hemostasis scan before committing to closure layers

4) First-Assist Post-Case Debrief Template

Use this after difficult cases, new procedures, or orientation weeks.

5-Minute Debrief

What went well?

- Exposure strengths
- Anticipation wins
- Communication that improved flow

Where did flow break down?

- Delay in instruments?
- Weak procedural knowledge?
- Missed preference cues?
- Poor field control?

What needs to improve before the next similar case?

- One technical adjustment

- One anticipation adjustment
- One setup adjustment

What should be standardized for the team?

- Preference cards
- Setup checklist
- Conduit prep sequence
- Closure routine
- Onboarding teaching points

5) Program-Level Competency Checklist

This section is useful for lead APPs, service line leaders, and educators building a repeatable onboarding process.

Mark each item as: **Observed / Needs Coaching / Independent / Teaches Others**

Case Preparation

- Reviews chart and indication before case
- Understands case sequence
- Identifies key risks and likely points of difficulty
- Arrives with correct mental model for surgeon and operation

OR Flow

- Sets up efficiently
- Maintains situational awareness during transitions
- Anticipates next steps accurately
- Communicates clearly with surgeon, scrub, and anesthesia when appropriate

Technical Assistance

- Provides effective exposure
- Handles tissue appropriately
- Supports hemostasis efficiently
- Assists with closure in organized, safe sequence

Professional Reliability

- Remains composed under pressure
- Accepts correction well
- Learns surgeon-specific preferences quickly
- Contributes to a calm room, not a chaotic one

Final Takeaway

The highest-value first assistants in cardiac surgery are not just technically capable. They are predictable, prepared, calm, and system-minded.

That matters because cardiac surgery teams do not scale well on heroics. They scale on standards.

If your onboarding depends on whoever happens to be teaching that day, you do not have a training system yet. You have tribal knowledge.

This template is meant to help fix that.

Want Help Building a Stronger Cardiac Surgery APP Training System?

Pol Martin D. Vedar, PA-C, MBA works with cardiac surgery teams on:

- APP onboarding and first-assist development
- Training systems and competency frameworks

- Retention-focused team design
- Clinical operations and quality improvement

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Educational resource only. Adapt to surgeon preference, institutional protocol, and scope-of-practice requirements.